

Bulan 4: Django Advanced Features

Minggu 1: Class-Based Views & Generic Views

Hari 1: Function vs Class-Based Views

Topik: View as class, `as_view()`, advantages of CBV

Proyek: CBVIntro

Intro ke Class-Based Views

- Convert function views ke CBV
- View class dengan `get()` dan `post()`
- `as_view()` dalam urls
- Compare dengan function views
- Teknologi: Class-Based Views basics

Hari 2: Generic Display Views

Topik: `ListView`, `DetailView`

Proyek: ProductListDetail

Product catalog dengan generic views

- `ListView` untuk semua products
- Pagination dengan `ListView`
- `DetailView` untuk product detail
- `template_name` customization
- `context_object_name`
- Teknologi: `ListView`, `DetailView`

Hari 3: Generic Editing Views

Topik: `CreateView`, `UpdateView`, `DeleteView`

Proyek: ArticleCRUD

CRUD artikel dengan generic views

- `CreateView` untuk create artikel
- `UpdateView` untuk edit
- `DeleteView` dengan confirmation

- success_url
- form_class specification
- Teknologi: CreateView, UpdateView, DeleteView

Hari 4: Mixins & View Customization

Topik: LoginRequiredMixin, UserPassesTestMixin, custom mixins

Proyek: SecureViews

Protected views dengan mixins

- LoginRequiredMixin untuk auth required
- UserPassesTestMixin untuk ownership check
- Custom mixin untuk logging
- Multiple mixins dalam satu view
- Teknologi: View mixins

Hari 5: FormView & TemplateView

Topik: FormView, TemplateView, RedirectView

Proyek: ContactSystem

Contact form dengan FormView

- FormView untuk contact form
- form_valid() override
- Send email (simulate)
- TemplateView untuk static pages
- Success message
- Teknologi: FormView, TemplateView

Hari 6: Pagination

Topik: Paginator, Page, page_obj in template

Proyek: BlogWithPagination

Blog dengan pagination

- Paginator object
- ListView dengan pagination

- Previous/Next links
- Page numbers
- Custom per_page
- Teknologi: Pagination

Hari 7: Review & Mini Project

Mini Project: NewsPortal

Portal berita dengan CBV

Features:

- ListView untuk news articles (paginated)
- DetailView untuk article detail
- CreateView untuk submit article (auth required)
- UpdateView untuk edit (author only)
- DeleteView untuk delete (author only)
- Category filter
- Search functionality
- UserPassesTestMixin untuk ownership
- Pagination
- Teknologi: Generic views, Mixins, Pagination

Minggu 2: Django REST Framework (DRF) Basics

Hari 8: REST & API Concepts Review

Topik: REST principles, JSON, API endpoints, HTTP methods

Proyek: APIPlanning

Plan REST API design

- Design endpoints untuk blog API
- Plan request/response formats
- Document API endpoints
- HTTP methods untuk CRUD
- Status codes mapping

Hari 9: DRF Installation & Setup

Topik: Install DRF, add to INSTALLED_APPS, basic API view

Proyek: DRFSetup

Setup Django REST Framework

- pip install djangorestframework
- Add 'rest_framework' to INSTALLED_APPS
- Create simple APIView
- Return JSON response
- Test dengan browser
- Teknologi: DRF setup

Hari 10: Serializers Basics

Topik: Serializer class, fields, serialization/deserialization

Proyek: ProductSerializer

Serializer untuk products

- Create ProductSerializer
- Serialize queryset
- Deserialize JSON to object
- Validation
- save() method
- Teknologi: DRF Serializers

Hari 11: ModelSerializers

Topik: ModelSerializer, Meta class, fields, read_only_fields

Proyek: BlogAPISerializer

Serializer untuk blog

- PostSerializer dengan ModelSerializer
- Meta class configuration
- Nested serializers (Category)
- read_only_fields
- Custom fields
- Teknologi: ModelSerializer

Hari 12: APIView & Function-Based Views

Topik: @api_view decorator, APIView class, Request, Response

Proyek: TaskAPIView

API views untuk tasks

- @api_view(['GET', 'POST'])
- Handle GET request (list)
- Handle POST request (create)
- APIView class untuk detail
- get(), post(), put(), delete() methods
- Teknologi: DRF views

Hari 13: Generic API Views

Topik: ListAPIView, RetrieveAPIView, CreateAPIView

Proyek: ProductAPIGeneric

Product API dengan generic views

- ListAPIView untuk list products
- RetrieveAPIView untuk detail
- CreateAPIView untuk create
- UpdateAPIView untuk update
- DestroyAPIView untuk delete
- Teknologi: DRF generic views

Hari 14: ViewSets & Routers

Topik: ModelViewSet, routers, automatic URL routing

Proyek: BlogAPIViewSet

Complete blog API dengan ViewSet

- PostViewSet extends ModelViewSet
- Auto CRUD endpoints
- DefaultRouter
- Automatic URL routing
- queryset & serializer_class

- Teknologi: ViewSets, Routers

Minggu 3: DRF Advanced & Authentication

Hari 15: Filtering & Searching

Topik: django-filter, SearchFilter, OrderingFilter

Proyek: ProductSearchAPI

API dengan filtering

- Install django-filter
- Add filter_backends
- SearchFilter untuk search
- OrderingFilter untuk sorting
- Custom FilterSet
- Teknologi: DRF filters

Hari 16: Pagination in DRF

Topik: PageNumberPagination, LimitOffsetPagination, CursorPagination

Proyek: PaginatedAPI

API dengan pagination

- Configure DEFAULT_PAGINATION_CLASS
- PageNumberPagination
- Custom page size
- LimitOffsetPagination
- Response format dengan pagination
- Teknologi: DRF pagination

Hari 17: Token Authentication

Topik: TokenAuthentication, obtain_auth_token, Token model

Proyek: TokenAuthAPI

API authentication dengan token

- Add rest_framework.authtoken
- Migrate Token model

- TokenAuthentication class
- /api/auth/login/ endpoint
- Return token
- Protected endpoints
- Teknologi: Token authentication

Hari 18: Permissions in DRF

Topik: IsAuthenticated, IsAuthenticatedOrReadOnly, custom permissions

Proyek: PermissionAPI

API dengan permissions

- IsAuthenticated untuk create
- IsAuthenticatedOrReadOnly untuk list
- Custom IsOwnerOrReadOnly
- Object-level permissions
- has_permission() & has_object_permission()
- Teknologi: DRF permissions

Hari 19: Nested Serializers & Relations

Topik: Nested serialization, depth, PrimaryKeyRelatedField, SlugRelatedField

Proyek: NestedBlogAPI

API dengan nested data

- PostSerializer dengan nested CategorySerializer
- CommentSerializer nested dalam Post
- depth option
- PrimaryKeyRelatedField untuk relations
- Write to nested serializers
- Teknologi: Nested serializers

Hari 20: Validation in DRF

Topik: Field-level validation, object-level validation, custom validators

Proyek: ValidatedProductAPI

API dengan comprehensive validation

- validate_fieldname() methods
- validate() untuk object-level
- Custom validator functions
- ValidationError responses
- Error messages
- Teknologi: DRF validation

Hari 21: Review & Mini Project

Mini Project: TaskManagerAPI

RESTful API untuk task management

Models:

- User (built-in)
- Project
- Task
- Comment

Endpoints:

- /api/auth/register/
- /api/auth/login/
- /api/projects/ (list, create)
- /api/projects/{id}/ (retrieve, update, delete)
- /api/tasks/ (list, create, filter by project, status)
- /api/tasks/{id}/ (retrieve, update, delete)
- /api/tasks/{id}/comments/ (list, create)

Features:

- Token authentication
- Permission (owner only untuk update/delete)
- Filtering (by project, status, priority)
- Search tasks
- Pagination
- Nested serializers (project dalam task)

- Validation (due_date must be future)
- Teknologi: DRF, ViewSets, Routers, Authentication, Permissions, Filtering

Minggu 4: Testing, Performance & Best Practices

Hari 22: Testing Django Views

Topik: TestCase, Client, test_get, test_post, assertions

Proyek: ViewTests

Test views Django

- Create test case
- Test GET request
- Test POST dengan data
- Test redirects
- Test context data
- assertEquals, assertContains
- Teknologi: Django testing

Hari 23: Testing Models & ORM

Topik: Model tests, database fixtures, setUp, tearDown

Proyek: ModelTests

Test models & queries

- Test model creation
- Test model methods
- Test QuerySet filtering
- Test model validations
- Test relationships
- Teknologi: Model testing

Hari 24: Testing DRF APIs

Topik: APITestCase, APIClient, test API endpoints

Proyek: APITests

Test REST API

- APITestCase class
- Test GET endpoint
- Test POST create
- Test authentication
- Test permissions
- status_code assertions
- Teknologi: DRF testing

Hari 25: Query Optimization

Topik: select_related, prefetch_related, only, defer, Django Debug Toolbar

Proyek: OptimizedQueries

Optimize database queries

- Install Django Debug Toolbar
- Identify N+1 queries
- Use select_related untuk ForeignKey
- Use prefetch_related untuk ManyToMany
- only() & defer() untuk fields
- Teknologi: Query optimization

Hari 26: Caching Basics

Topik: Cache framework, per-view caching, template fragment caching

Proyek: CachedBlog

Blog dengan caching

- Configure cache backend (dummy/local memory)
- @cache_page decorator
- Cache QuerySet results
- Template fragment caching
- Cache invalidation
- Teknologi: Django caching

Hari 27: Signals & Background Tasks

Topik: Django signals, post_save, pre_save, receivers

Proyek: SignalSystem

Automated actions dengan signals

- post_save signal untuk create profile
- pre_save untuk auto-slug
- Custom signal
- Signal receivers
- Disconnect signals
- Teknologi: Django signals

Hari 28: Review & Final Project Bulan 4

Final Project: EcommerceAPI

E-commerce REST API lengkap

Models:

- User, UserProfile
- Category
- Product
- Cart, CartItem
- Order, OrderItem
- Review

API Endpoints:

- Auth: register, login, profile
- Products: CRUD, filter by category, search, reviews
- Cart: add to cart, update quantity, remove, clear
- Orders: create from cart, list orders, order detail
- Reviews: create, list by product

Features:

- Token authentication
- Permission system (seller vs buyer)
- Filtering & search products
- Pagination

- Nested serializers
 - Cart management logic
 - Order creation dari cart
 - Stock validation
 - Average rating calculation
 - Image upload untuk products
 - Comprehensive tests (50+ test cases)
 - Query optimization (select_related, prefetch_related)
 - API documentation (manual)
 - Teknologi: DRF, Authentication, Permissions, Signals, Testing, Optimization
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Progress Check Bulan 4: ☒ Class-Based Views & Generic Views ☒ View Mixins ☒ Django REST Framework fundamentals ☒ Serializers & ModelSerializers ☒ ViewSets & Routers ☒ DRF Authentication (Token) ☒ DRF Permissions ☒ Filtering & Pagination ☒ Testing (views, models, APIs) ☒ Query optimization ☒ Caching basics ☒ Django signals

Total Proyek: 60+ proyek & 3 major API projects **Siap lanjut:** Bulan 5 - Advanced DRF & Production Concepts